

OKAYAMA, Japan

WMO: 477680

Lat: 34.6858N

Lon: 133.9253E

Elev: 23

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 29.7	(c) 31.5	(d) 16.8	(e) 12.8	(f) 39.6	(g) 19.2	(h) 14.5	(i) 40.3	(j) 27.1	(k) 40.9	(l) 24.0	(m) 41.7	(n) 6.5	(o) 270	(p) 0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.2	(c) 94.3	(d) 77.7	(e) 92.4	(f) 77.2	(g) 90.6	(h) 76.7	(i) 79.4	(j) 89.7	(k) 78.6	(l) 88.3	(m) 77.9	(n) 87.1	(o) 9.4	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.9	140.1	83.4	76.1	136.3	83.3	75.3	132.7	83.2	42.9	90.1	42.1	88.4	41.4	87.2	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.7	(b) 18.5	(c) 15.7	(e) 26.3	(f) 97.1	(g) 2.2	(h) 1.8	(i) 24.7	(j) 98.4	(k) 23.4	(l) 99.4	(m) 22.2	(n) 100.4	(o) 20.6	(p) 101.7		
(a) 21.7	(b) 18.5	(c) 15.7	(e) 24.0	(f) 80.8	(g) 2.2	(h) 1.2	(i) 22.4	(j) 81.7	(k) 21.1	(l) 82.4	(m) 19.8	(n) 83.0	(o) 18.2	(p) 83.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	41.1	42.4	48.3	58.3	67.7	74.4	82.0	84.1	76.6	65.9	54.8	45.3
	DBStd	15.60	3.83	4.68	5.37	5.76	4.24	4.20	4.11	3.25	5.08	5.27	5.50	4.65
	HDD50	775	276	218	99	7	0	0	0	0	0	0	16	159
	HDD65	3096	741	632	518	210	23	0	0	0	0	52	308	612
	CDD50	5099	1	6	46	255	549	731	993	1057	797	492	160	12
	CDD65	1942	0	0	0	8	106	281	528	592	347	79	1	0
	CDH74	18744	0	0	0	58	611	1863	5769	7229	2909	305	0	0
	CDH80	7181	0	0	0	3	87	476	2406	3304	876	29	0	0
(14) Wind	WSAvg	6.6	7.3	6.9	7.1	6.9	6.6	6.3	6.4	6.6	6.2	5.6	5.7	6.9
(15) Precipitation	PrecAvg	46.3	1.5	2.0	3.3	3.5	4.7	6.7	6.6	4.1	6.0	3.7	2.5	1.8
	PrecMax	64.8	4.4	4.1	7.7	7.4	12.0	15.2	14.6	12.0	17.6	10.6	5.9	4.1
	PrecMin	31.1	0.2	0.6	1.3	1.2	0.9	0.9	1.6	0.7	2.0	1.0	0.5	0.2
	PrecStd	9.3	1.0	0.9	1.4	1.5	2.4	3.2	3.5	3.0	3.9	2.5	1.5	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	61.8	69.2	79.1	85.8	91.6	96.1	97.0	92.7	83.3	71.8	61.3
		MCWB	47.6	53.3	54.7	60.5	64.4	74.6	78.5	78.4	76.7	71.6	61.6	53.3
	2%	DB	53.1	57.1	65.1	75.3	82.4	87.3	94.0	94.9	89.5	79.6	69.2	58.8
		MCWB	45.5	49.0	53.0	58.9	63.6	72.1	77.7	77.9	75.5	68.5	60.0	51.8
	5%	DB	51.0	54.1	61.7	72.0	79.8	84.8	92.1	93.1	87.2	77.3	66.9	56.4
		MCWB	43.7	46.4	51.3	57.8	62.8	70.3	77.3	77.5	74.4	66.5	58.6	49.0
	10%	DB	48.9	51.3	58.6	69.1	77.5	82.6	90.1	91.5	85.0	75.1	64.4	53.9
		MCWB	42.0	44.1	49.7	56.3	62.4	69.4	76.8	77.0	73.3	64.7	56.7	46.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	55.9	58.7	66.4	71.2	77.0	80.5	80.7	78.8	74.8	65.6	57.5
		MCDB	53.9	60.7	63.6	72.1	76.8	86.2	92.6	92.2	87.9	80.1	68.1	59.7
	2%	WB	47.0	51.0	56.1	63.5	68.7	75.4	79.2	79.7	77.4	71.5	62.7	52.9
		MCDB	50.9	55.3	61.2	68.8	74.7	83.3	89.7	90.2	85.3	77.7	66.8	56.7
	5%	WB	44.9	47.9	53.7	61.2	66.9	74.1	78.5	78.9	76.5	69.1	60.4	50.1
		MCDB	49.7	52.3	59.6	67.9	73.7	81.2	88.2	88.8	84.0	74.8	65.1	54.6
	10%	WB	42.7	45.0	50.9	59.2	65.3	72.6	77.7	78.2	75.5	66.5	58.0	47.9
		MCDB	48.1	50.2	57.6	66.2	72.9	79.0	86.9	87.6	82.8	72.9	63.1	53.0
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	14.3	16.0	17.0	16.8	13.5	12.7	13.2	12.9	14.2	14.3	13.7
		MCDBR	16.0	17.4	20.1	21.4	21.3	17.1	15.6	15.5	14.0	15.3	15.4	16.0
		MCWBR	10.4	11.4	11.9	9.7	8.5	5.9	4.6	4.7	5.3	7.2	9.0	10.3
	5% WB	MCDBR	14.7	14.9	16.4	15.2	14.5	12.4	14.3	13.9	12.1	11.9	12.7	14.0
		MCWBR	10.8	11.4	12.3	9.2	8.2	5.1	4.7	4.5	5.1	7.5	9.0	10.5
(40) Clear-Sky Solar Irradiance	taub		0.416	0.461	0.528	0.557	0.559	0.581	0.581	0.552	0.503	0.457	0.432	0.409
	taud		2.086	1.987	1.844	1.831	1.887	1.908	1.961	2.028	2.126	2.163	2.133	2.124
	Ebn at Noon		239	242	236	237	239	233	232	236	241	241	233	233
	Edh at Noon		40	49	62	66	63	62	58	54	46	41	38	37
(44) All-Sky Solar Radiation	RadAvg		721	921	1241	1534	1679	1522	1599	1648	1282	1056	781	642
	RadStd		49	76	95	134	155	134	213	168	125	101	65	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page